A3: Process Capability Jupyter Analysis

Context and Rationale

Manufacturing Setting: The Ubuntu Naturals Port Louis liquid-fill line (Mauritius, Tier 2; Stage 4) fills **Spirulina Power** bottles, and watches two things while it does: the pasteurisation temperature ahead of the filler, and the volume that ends up in each bottle. Until recently both were checked on paper. The line is now instrumented -- IIoT sensors logging temperature at 1-second intervals, part of the group's Port Louis digital-twin pilot -- and it is moving onto automated IIoT-based statistical monitoring, with the readings piling up faster than anyone is reading them.

Rationale: A pasteuriser running cool does not announce itself. The consequence turns up later as microbial contamination, and after that as a product recall, consumer harm, or a shutdown ordered by PPB, EFDA, EFSA or FDA. Fill volume is less dramatic and no cheaper: underfill is a regulatory breach, overfill is margin poured away a few millilitres at a time. What has changed at Port Louis is the volume of evidence -- the new IIoT sensors produce more readings in an hour than the paper checks collected in a month, and nobody can read that by eye. A notebook can. Yours has to give the Ubuntu quality manager two things: the evidence-based case for automating statistical monitoring, and a working prototype of the anomaly detection it would run.

Brief

You are given two datasets from the Port Louis line: 10,000 pasteurisation temperature readings logged a second apart, in `A03_port_louis_pasteurisation_temp_10000.csv`, and 200 subgroups of n=5 fill volume measurements, in `A03_spirulina_fill_volume_200x5.csv`. Explore both, test whether they are normal, and build the control charts in code rather than by clicking. Calculate the capability indices for the fill process. Then write a simple anomaly detection model for the temperature series -- something that would flag a potential equipment fault while there is still time to do something about it.

Key Requirements

- Explore both datasets -- descriptive statistics, histograms, box plots -- and say in markdown what each one tells you
- Test for normality by at least two methods (Shapiro-Wilk and a Q-Q plot, say), and work out what it does to your SPC limits if the data is not normal
- Build X-bar/R charts for the fill volume data and an Individuals/Moving Range chart for the temperature data, in Python (matplotlib, plotly, or equivalent)

- Calculate Cp, Cpk, Pp and Ppk for the fill volume process, and read the numbers back in terms of product safety and export-bound liquid rather than as statistics
- Implement one anomaly detection approach for the temperature time-series -- a rolling z-score, CUSUM, or an isolation forest -- that could act as an early warning for pasteuriser fouling or sensor drift
- Close with a markdown section of recommendations for the plant quality manager, in language someone with no statistics background can act on

Production Guidelines

1. **Set up the notebook environment**: Import the libraries you need (pandas, numpy, scipy, matplotlib/plotly, and scikit-learn if you take the isolation forest route). Load both datasets attached to this assignment -- `A03_port_louis_pasteurisation_temp_10000.csv`, 10,000 one-second pasteurisation temperature readings with the specification limits stated in its header, and `A03_spirulina_fill_volume_200x5.csv`, 200 subgroups of 5 fill volumes with its target and limits in the header. Confirm the shapes, types and completeness before you trust either of them.
2. **Perform exploratory data analysis**: For both datasets, compute the descriptive statistics -- mean, median, standard deviation, range, skewness, kurtosis. Plot histograms, box plots and the time series. Then write the markdown cells that say what each plot shows about the distribution, and about any pattern visible in it.
3. **Test for normality**: Apply at least two methods -- the Shapiro-Wilk test (or Anderson-Darling for large samples) and a Q-Q plot. Then answer what follows from the result: if the temperature data is non-normal, what happens to standard SPC limits applied to it, and what transformation or non-parametric approach would you reach for instead?
4. **Construct control charts programmatically**: Build X-bar/R charts for fill volume, which is subgrouped, and Individuals/Moving Range charts for temperature, which is not. Calculate the control limits from the data yourself, using the appropriate constants and formulae. Do not call a pre-built SPC library and take its word for the answer. One decision comes before the arithmetic: which data the limits are fitted on. Limits are meant to describe a process while it is in control, so fit them on a stretch of the temperature series where nothing has started moving yet -- the run opens with one -- and then apply those limits across the whole series to see what the rest of the run does against them. Fit to all 10,000 readings at once as well, and compare: what happens later in the run drags the centre line with it, and the quiet opening stretch is what ends up flagged. Say which set you kept and why. Control limits describe a process that is holding steady, and the temperature run does not hold steady for all 10,000 readings, so the window you fit on is a decision you have to make and defend rather than a detail of the arithmetic.
5. **Calculate and interpret capability indices**: For the fill volume process, compute Cp, Cpk, Pp, and Ppk against the specification limits. Then read the numbers back in the product safety / export context: what does a Cpk of, say, 0.9 mean for how often a Spirulina Power bottle leaves Port Louis underfilled or overfilled?
6. **Build an anomaly detection model**: Implement at least one approach for the temperature time-series -- a rolling z-score with a 60-second window, a CUSUM chart, or an isolation forest. Explain the method, tune the sensitivity threshold rather than accepting a default, and mark the

anomalies you detect on a time-series plot. Then say how this would flag pasteuriser fouling or sensor drift while there is still time to act, before a safety breach occurs.

7. **Write the recommendations section**: Close with a markdown section for the Port Louis quality manager, in language that assumes no statistics background. Name the actions you want taken -- process adjustments, monitoring thresholds, equipment checks -- and say what automated monitoring gives the plant that the previous paper-based system did not.

Marking Scheme

Criterion	Weight	Fail (0-39)	Pass (40-59)	Merit (60-69)	Distinction (70-84)	Stretch (85-100)
Exploratory Data Analysis	20%	Minimal or no EDA; statistics missing or incorrect; visualisations absent or uninterpretable.	Basic descriptive statistics computed for both datasets; at least one visualisation per dataset; markdown narrative present but superficial.	The full set of descriptive statistics with well-formatted visualisations (histograms, box plots, time-series); markdown narrative explains patterns, outliers, and distributional shape for both datasets.	EDA reveals non-obvious insights (e.g., time-of-day effects in temperature, subgroup-to-subgroup trends in fill volume); visualisations are multi-panel, well-labelled, and publication-ready.	EDA includes correlation analysis between variables, autocorrelation analysis for the time-series, and decomposition of trends/seasonality; narrative proposes hypotheses about process behaviour to be tested in subsequent sections.
Normality Testing	15%	No normality testing, or tests applied incorrectly with wrong conclusions.	At least one normality test applied correctly with Q-Q plot; basic statement of pass/fail for normality assumption.	Two tests applied correctly; Q-Q plots well-constructed; clear discussion of implications for SPC (e.g., "non-normality means standard 3-sigma limits may not achieve the expected 99.73% coverage").	Discussion extends to practical impact: quantifies the actual false alarm rate under the observed distribution; proposes alternatives (e.g., Box-Cox transformation, percentile-based limits, or non-parametric charts).	Compares multiple transformation approaches, showing before/after Q-Q plots; discusses robustness of SPC methods to the specific type of non-normality observed; references food industry guidance on handling non-normal process data.

Criterion	Weight	Fail (0-39)	Pass (40-59)	Merit (60-69)	Distinction (70-84)	Stretch (85-100)
Control Chart Construction	20%	Charts not constructed, or constructed with incorrect formulae; no distinction between the two chart types needed.	Both chart types present (X-bar/R for fill volume, I/MR for temperature); control limits calculated but may have minor errors; basic visual presentation.	Charts correctly constructed with visible formulae and clear annotations; out-of-control points identified; appropriate chart types justified for each dataset; the temperature limits state which stretch of the series they were fitted on, and both fits are compared with a stated choice of which set was kept.	Charts include zone markings (1-sigma, 2-sigma, 3-sigma); code is well-structured, modular, and reusable; chart interpretation is specific to the Spirulina Power liquid-fill context; the baseline-window choice is justified rather than asserted -- limits fitted across all 10,000 readings absorb the later movement into the centre line and widen with it, so the drift stops tripping them and the settled opening stretch is what gets flagged instead.	Charts are interactive (plotly with hover-over data); code includes functions that could be deployed as a reusable monitoring module; includes sensitivity analysis varying subgroup size or control limit width.
Capability Analysis	15%	Capability indices not calculated, or fundamentally incorrect.	Cp and Cpk calculated correctly for fill volume; basic interpretation of whether the process is capable.	All four indices calculated; clear explanation of capability vs. performance distinction; results interpreted with reference to product safety (underfill risk) and commercial impact (overfill cost).	Includes confidence intervals for the indices; relates Cpk to expected defect rate (PPM); discusses the practical meaning of the results for the Port Louis quality manager's daily decision-making.	Proposes a target Cpk and simulates the process changes needed to achieve it; includes a yield/loss analysis linking Cpk improvement to financial outcomes; discusses how real-time capability monitoring could be implemented.

Criterion	Weight	Fail (0-39)	Pass (40-59)	Merit (60-69)	Distinction (70-84)	Stretch (85-100)
Anomaly Detection	20%	No anomaly detection attempted, or method is fundamentally flawed.	One method implemented with default parameters; some anomalies detected and shown on a plot; brief explanation of the method.	Method well-explained with parameter tuning justified; detected anomalies plotted clearly on the time-series; discussion of how the method would work as an early warning system for pasteuriser faults.	Threshold tuning is evidence-based (e.g., using a validation set or false positive rate analysis); comparison of method sensitivity vs. specificity; practical deployment considerations discussed (latency, computational cost).	Multiple methods compared (e.g., rolling z-score vs. CUSUM vs. isolation forest) with a quantitative comparison of detection performance; includes a proposed alerting framework with escalation levels (warning, action, shutdown) linked to product safety risk.
Code Quality and Narrative	10%	Code does not run; no markdown narrative; poor or absent structure.	Code runs end-to-end; some markdown cells but narrative is minimal; variable names are unclear.	Well-structured notebook with clear markdown sections; code is commented; narrative connects each analysis step to the Port Louis liquid-fill context.	Code follows best practices (functions, clear naming, no repeated code blocks); narrative reads as a coherent analytical report; visualisations have consistent styling.	Notebook is structured for reproducibility (requirements cell, random seeds, parameterised functions); includes a table of contents; could serve as a template for analysing other production lines across Ubuntu Naturals.

Stretch Goal

An outstanding submission would go further and build a prototype real-time monitoring dashboard inside the notebook, with ipywidgets or Panel, simulating how incoming temperature data would be processed, charted and flagged on a live line. It would compare several anomaly detection algorithms on numbers rather than on description, propose tiered alerting -- operator warning, supervisor alert, automatic line shutdown -- tied to product safety risk levels, and cost automated monitoring against the paper-based system with the IIoT sensor investment Port Louis has already made on the books.

Submitting This Artefact

What you hand in: Annotated Jupyter notebook (.ipynb) with Python code, statistical analysis, visualisations, and markdown narrative.

This is a Phase 2 (Applied) artefact. It builds on A2 (SPC foundations), and the work you do here carries into A8 (data pipeline skills) -- so it is worth doing properly the first time rather than twice.

Three files are attached to the assignment: this brief, as `A03_process_capability_jupyter_analysis.pdf`, and the datasets it refers to, as `A03_port_louis_pasteurisation_temp_10000.csv` and `A03_spirulina_fill_volume_200x5.csv`. A worked exemplar from another Ubuntu Naturals site is attached too, with the marker's note on it, and `A03_genai_guide.pdf` shows how to build your own submission with a free Claude account.

Name the file `A03_Process_Capability_[YourSurname].ipynb`, with your own surname in place of `[YourSurname]` -- the portfolio guide uses this convention throughout, and a marker looking for your work will look for that name.

Whatever else it contains, the file needs markdown cells explaining your reasoning between code cells; not just code output.